A Low-Cost and Low-Complexity Motion Sensor System Using IMU and Video-Based Tracking

CM2016 VT26 Project Carrier Course for Medical Engineers, Part 2 (60644)

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Abstract

Abstract to go here.

Contributions

Contributions so far have been shared equally between all group members. This includes the development of the web application codebase and the preparation and delivery of the group presentation.

For the progress report, Jack Hutton wrote the background information for the IMU's and Sensor Fusion (Section 2.4 and 2.6), the Sensor Fusion Methods (Section 3.3), the results for the Sensor Fusion and IMU Angles (Section 4.3 and 4.4) and the plan for future experiments (Section 4.5)

Alexander Jonsson wrote the sections concerning Video pose estimation (2.3), Signal Processing (background, 2.5), Synchronization (2.7), Signal Processing (method, 3.3) and IMU Filtering (4.2).

Sahand Khodabandehloo wrote the Introduction, as well as the Background (Sections 2.1 and 2.2), the Methods (Sections 3.1 and 3.2), and the Results (4.1), with a particular focus on the overall web application design.

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1 Introduction

1.1 Motivation

In competitive sports, small margins often determine the difference between success and failure. Improvements measured in milliseconds, degrees of joint angle, or subtle changes in coordination can significantly influence performance outcomes in athletic activities (e.g., sprinting, jumping, throwing, and strength training). Consequently, biomechanical analysis has become an increasingly important component of modern sports science, supporting evidence-based coaching, performance optimisation, and injury prevention (1).

Despite substantial scientific knowledge regarding movement quality and its impact on performance and injury risk, many training decisions continue to rely heavily on visual assessment and subjective coaching experience rather than objective biomechanical data (1). While expert observation remains valuable, it is inherently limited in precision and reproducibility. This gap highlights the need for practical tools that make quantitative movement analysis more accessible in everyday training environments.

1.2 Research objectives

To address this need, a browser-based motion tracking application, *MoveSync*, was developed within this project. The objectives of the system are to:

1. Perform browser-based video pose estimation to extract 2D skeletal keypoints from recorded sports movements.
2. Develop a synchronisation method that aligns video frames with IMU data despite independent time bases.
3. Provide an intuitive interface for side-by-side visualisation of synchronised video and sensor data to support performance analysis.

The system is specifically designed for offline analysis of recorded sports movements, prioritising interpretability, accessibility, and usability in real-world training contexts. A link to the MoveSync application will be publicly available in a GitHub repository.

1.3 Aim and research question

The aim of this project is to design a low-cost, low-complexity, browser-based motion tracking prototype tailored for sports performance analysis. The MoveSync system should enable

athletes and coaches to analyse recorded movements using only a consumer-grade camera and wearable IMU sensors, without requiring specialised laboratory equipment. This leads to the following research question:

> How can video-based pose estimation and IMU data be integrated into a single browser-based application to support accessible, practical, and reproducible sports performance analysis?

1.4 Requirements

To fulfil its intended role in sports performance settings, the system should meet the following requirements:

- Be affordable and rely on commercially available cameras and wearable IMUs.
- Operate entirely in a standard web browser without complex installation or specialised software.
- Support straightforward import of recorded sports videos and IMU CSV data.
- Provide synchronised visualisation of skeletal motion and sensor signals.
- Allow event marking (e.g., jump take-off, landing, sprint start, lift initiation) and annotation for structured performance review.
- Enable export of session data and reports for documentation and coach–athlete feedback.

The system is intended for exploratory performance analysis and training support rather than laboratory-grade biomechanical validation. It should NOT be used to establish a diagnosis or medical evaluation.

1.5 Report structure

The report first introduces the motivation and objectives for developing an accessible sports performance motion analysis tool. It then presents background information on motion tracking principles, IMUs, video pose estimation, and synchronisation methods. Subsequently, the system design and implementation of the MoveSync browser-based application are described. The results section demonstrates the functionality of the prototype in sports-related test scenarios. Finally, the discussion addresses system performance, limitations, ethical considerations related to athlete data handling, and directions for future development.

2 Background

2.1 Motion capture techniques

Motion capture (MoCap) refers to technologies that record and quantify movement for biomechanical analysis. By transforming movement into measurable information, in terms of kinematic and dynamic variables (e.g., joint trajectories, segment velocities, accelerations, and event timing), movement patterns can be evaluated systematically and reproducibly.

Laboratory-based optical systems enable high-accuracy three-dimensional reconstruction of joint kinematics and are widely used in research settings (1). However, their use is typically restricted to controlled environments due to technical complexity and infrastructure requirements.

Wearable Inertial Measurement Units (IMUs) offer a portable alternative by measuring linear acceleration and angular velocity to estimate segment motion and joint kinematics (2, 3). IMU-based systems support data collection in field environments and have become increasingly common in sports science and injury monitoring (4).

More recently, advances in computer vision have enabled markerless, video-based pose estimation using standard RGB cameras (5, 6). These approaches reduce setup complexity but may be sensitive to camera perspective, occlusion, and two-dimensional reconstruction constraints.

Each technique involves trade-offs between measurement accuracy, precision, fidelity, portability, and practical deployment. In sports performance contexts, combining wearable inertial sensing with video-based pose estimation offers a promising compromise between measurement capability and real-world applicability.

2.2 Motion tracking in sports performance

In sports performance settings, motion tracking serves a different primary objective than in laboratory biomechanics research: it must support training decisions in dynamic, time-constrained, and often non-controlled environments. Coaches and athletes require feedback that is not only accurate but also rapid, interpretable, and practically deployable during regular training sessions. Furthermore, motion capture enables objective assessment of technique, monitoring of training adaptations over time, identification of asymmetries, and evaluation of movement efficiency.

Quantitative motion analysis enables objective assessment of technique, monitoring of performance progression, detection of movement asymmetries, and evaluation of efficiency (1).

Such information can inform load management, technical adjustments, and injury prevention strategies (4). However, in everyday practice, access to laboratory-grade systems is limited, and decision-making frequently relies on visual inspection and subjective judgment.

As a result, there is a growing demand for motion analysis tools that balance measurement capability with usability and accessibility. Systems that integrate portable sensing technologies with intuitive analysis interfaces may help bridge the gap between biomechanical research and applied sports performance practice.

2.3 Video pose estimation

Video pose estimation is a computer-vision approach that infers a human pose from standard color (RGB) video frames by predicting a set of body keypoints for each frame, which can be visualized as a skeletal overlay (5). In this project, pose estimation was performed in the browser using TensorFlow.js models. The two detectors considered (PoseNet and MoveNet) are compared in Table 1 (5, 6).

2.3.1 PoseNet

PoseNet is an established pose-estimation model that was made widely accessible through TensorFlow.js (5). In this work, PoseNet was primarily used as a baseline to contextualize model choices and to provide a well-documented reference implementation (Table 1) (5).

2.3.2 MoveNet

MoveNet is a newer detector provided via the TensorFlow.js pose-detection API (6). The model and its surrounding pipeline are designed to improve robustness in video, including measures intended to reduce temporal instability (e.g., jitter) (6). In this implementation, the detector is instantiated once (e.g., `createDetector`) and pose inference is run per frame (e.g., `estimatePoses`), returning 2D keypoints with associated confidence scores (6).

MoveNet offers two single-person variants, Lightning and Thunder. Their trade-offs and the resulting choice for the uploaded-video workflow are summarized in Table 2 (6). Since the final system focuses exclusively on offline processing of uploaded videos, the selection prioritizes keypoint stability and precision over inference latency.

Table 1: Comparison of PoseNet and MoveNet in the context of this project.

Aspect	**PoseNet**	**MoveNet (SinglePose)**
Core output	2D body keypoints (x,y) + confidence per frame	2D body keypoints (x,y) + confidence per frame
Keypoints	17 keypoints (COCO-style)	17 keypoints (COCO-style)
Typical purpose	General-purpose baseline, good for prototyping	efficient inference + stable tracking
Strengths	Well-known, highly configurable; good documentation	Faster and more stable on video; good for browser-based use
Weaknesses	Slower at higher accuracy settings; less stable in fast motion	Fewer classic tuning knobs; mainly choose model variant
Main tunable parameters	`outputStride`, `inputResolution`, `multiplier` (trade speed vs accuracy)	Model variant (Lightning/Thunder), internal tracking/cropping behaviour
Best fit in our app	Baseline / comparison model	Primary model for uploaded-video analysis
Why we chose it	Reference model; exposes classic speed–accuracy trade-offs	Better keypoint reliability for practical video analysis

2.4 Inertial Measurement Units

Inertial Measurement Units (IMUs) are electronic devices that contain an array of sensors designed to measure motion. They typically comprise a combination of gyroscopes, accelerometers, and sometimes magnetometers to capture the rotational and translational motion of a body. The gyroscope measures angular velocity about each axis, the accelerometer measures linear acceleration along each axis, and the magnetometer provides a reference to the Earth's magnetic field for estimating absolute heading (7). Together, these sensors provide up to nine degrees of freedom (DOF) of raw motion data, which can be used to estimate the full three-dimensional orientation of a body segment (8).

Most IMU devices are micro-electro-mechanical systems (MEMS) which range in size and complexity. These devices typically operate on frequencies ranging from 100–2000 Hz depending on the sensor. MEMS IMUs are commonly used in wearable devices and consumer

Table 2: MoveNet variants considered for the uploaded-video workflow.

Aspect	**MoveNet Lightning**	**MoveNet Thunder**
Primary goal	Low latency / speed	Higher accuracy / stability
Typical input size (reported)	192×192	256×256
Keypoint quality (qualitative)	Good, but more jitter/less precise in challenging frames	Better precision and more stable keypoints (esp. occlusions / fast motion)
Runtime (qualitative)	Faster inference	Slower inference
Relevance for uploaded video	Speed is helpful, but not the main constraint	Accuracy/stability is prioritized because processing is offline
Decision in our project	Considered, but not selected	Selected for upload to improve keypoint reliability for analysis

electronics due to their small form factors, minimal power consumption, and low production cost (8). Higher performance IMUs do exist that use fibre optic gyroscopes (FOGs) or ring laser gyroscopes (RLGs), particularly in the maritime and aerospace fields, but these are significantly more expensive and complex (8).

IMU devices are becoming increasingly common in human motion analysis for biomechanical and clinical research. This is mainly because of their ability to measure movement outside of a laboratory setting whilst being low cost and small. This has significant applications in human health, physical recovery, and sports analysis (9). For sports biomechanics, kinematic data from IMUs can aid in both activity recognition and force estimation, both of which are critical for sports analysis (10).

There are several limitations to IMU technology, particularly MEMS IMUs, that constrain their use without additional correction or processing. MEMS IMUs are susceptible to a number of error sources including temperature-dependent drift, scale factor errors, cross-axis sensitivity, random measurement noise, and bias (11). Particularly significant issues arise as a result of noisy gyroscope and acceleration data which, when integrated over time, can cause cumulative orientation and position error. This necessitates signal processing to obtain more stable and accurate results for analysis.

2.5 Signal Processing

To acquire reliable measurements, it is important that appropriate filtering is applied to the signals. This is particularly important for acceleration data, since many features of interest, such as velocity and displacement, are obtained through single or double integration of acceleration. Because integration accumulates errors over time, even small measurement errors in the form of high-frequency noise or low-frequency bias in the acceleration signal can result in significant drift in the estimated velocity and position. Therefore, preprocessing and filtering of acceleration data are essential before integration (12).

Several approaches for filtering sensor data have been proposed in the literature. For example, a digital low-pass filter can be used to filter out high frequency noise, and a digital high-pass filter can be used to reduce offset errors and biases such as gravity (13). Other, more advanced approaches include sensor fusion algorithms, where inertial measurements are combined with complementary sensors to reduce drift and improve overall estimation accuracy (14)

2.6 Sensor Fusion

Sensor fusion is the process of combining data from multiple sensors to produce a more accurate and reliable estimate than any of the sensors could individually. For IMUs, sensor fusion can help reconcile the complementary error characteristics of the gyroscope, accelerometer, and magnetometer (15). Several approaches exist for fusing raw IMU sensor data into a stable orientation estimate (16).

The complementary filter is the simplest, splitting the frequency spectrum so that low-frequency accelerometer data and high-frequency gyroscope data are combined via complementary high-pass and low-pass filters. This method is computationally cheap but can be limited in accuracy (17). The Kalman filter is a similar but more sophisticated approach, as it models system and measurement noise to produce an optimal state estimate. It is widely used and can be highly accurate, but has significant computational requirements as well as requiring knowledge of the noise distribution (17). The Madgwick algorithm, developed by Sebastian Madgwick in 2010, offers a middle ground between the two and is widely used due to its computational efficiency.

The Madgwick algorithm represents orientation as a quaternion, using accelerometer and magnetometer data in an analytically derived gradient-descent algorithm to calculate the direction of the gyroscope measurement error (18). Quaternions are an alternative way of describing orientation in three-dimensional space using an ordered set of four numbers,

avoiding the gimbal lock associated with Euler angle systems (19). A single gain parameter, β, controls the balance between gyroscope integration and the correction from the accelerometer and magnetometer, enabling the filter to be tuned according to the relative confidence placed in each sensor (20).

After stable orientation estimates are obtained via sensor fusion, a wide range of further analyses become possible. Orientation can be expressed as Euler angles (roll, pitch, yaw) or a rotation matrix, which can be used to create a real-time 3D visualisation of the sensor's position and orientation. Furthermore, when multiple IMUs are worn on adjacent body segments simultaneously, the relative orientation between segments can be computed from their respective rotation matrices, enabling continuous joint angle estimation (21).

2.7 Synchronization

When combining video-based motion tracking with IMU recordings, both modalities typically follow independent time bases. Synchronization, therefore, refers to establishing a consistent mapping between the video timeline and the IMU timeline such that corresponding samples represent the same physical instant. A common and practical approach is event-based synchronization, where a distinct, clearly identifiable motion event (e.g., a hand clap, jump landing, or sudden knee flexion) is used as a reference point in both signals.

Let t_{Video} denote the timestamp of the reference event in the video and t_{IMU} the corresponding timestamp in the IMU recording. The temporal alignment can then be expressed as a constant time offset $\Delta t = t_{Video} - t_{IMU}$. Using this offset, any IMU sample at time t_{IMU} can be mapped to the video time. This assumption is appropriate when the dominant discrepancy between both data streams is a fixed start-time difference; more complex effects, such as clock drift, would require additional reference points or a more advanced alignment model.

Once synchronized, the IMU signals can be interpreted directly in the context of the observed movement in the video. This improves interpretability by linking quantitative measurements (e.g., accelerations and angular velocities, and potentially derived kinematic variables) to the corresponding visual motion, which supports clearer analysis and reduces ambiguity during inspection and annotation.

3 Methods

3.1 System Architecture

3.1.1 Architecture Design Principles

The MoveSync system was implemented using a modular client–server architecture based on JavaScript and HTML. The architectural design follows a functional separation of concerns, in which data acquisition, processing, synchronization, storage, and visualization are encapsulated in logically distinct modules. This modular decomposition minimizes interdependencies between components, thereby facilitating maintainability, scalability, and extensibility. Individual modules can be modified, replaced, or extended without directly affecting the overall system functionality. Furthermore, this separation also simplifies debugging, testing, and future feature integrations.

3.1.2 Session-Based Data Model

Within the system, a *session* is defined as a complete data import consisting of one video recording and its associated IMU data files (e.g., one video file and multiple IMU CSV files). Each session constitutes a unified analytical entity, serving as the fundamental unit for storage, synchronization, retrieval, and comparative analysis. This session-oriented data model ensures structural consistency across datasets, enabling systematic organization within the application. It also provides a clear distinction between raw data input and higher-level analysis functionalities.

3.1.3 Data Stream Processing and Synchronization

Video-based pose estimation data and IMU sensor measurements are handled as independent data streams. Each stream undergoes individual preprocessing, including parsing, formatting, and validation.

Temporal alignment, or time synchronisation, is subsequently performed to ensure coherent integration of multimodal motion data. By maintaining independent preprocessing pipelines while applying a unified synchronization stage, the system preserves flexibility in handling raw data sources while enabling combined analysis and visualization.

3.1.4 Client-Side Execution Environment

All computational processes are executed locally within a standard web browser environment. Video handling, IMU CSV parsing, synchronization algorithms, and visualization routines are performed entirely on the client side. This design eliminates the need for specialized software installation or dedicated server infrastructure. Consequently, the system can operate on commonly available hardware platforms, improving accessibility and portability.

3.2 Web Application Design and Implementation

The web application was developed as a centralized platform for managing, visualizing, and analyzing motion tracking sessions. An iterative development methodology was adopted, allowing incremental refinement of functionality, user interaction, and system performance throughout the implementation process. This approach enabled continuous validation of design decisions and facilitated the integration of new features without compromising architectural consistency. The application is organized into five primary functional views:

- **Dashboard**

  Provides an overview of recent sessions and general system statistics, offering users immediate insight into stored data and system status.

- **Upload**

  Enables the import of motion capture data, including video files and IMU CSV recordings. This module handles input validation and session initialization.

- **Library**

  Serves as a session management interface, allowing users to organize, retrieve, and manage previously imported datasets.

- **Session Viewer**

  Supports interactive visualization of synchronized video and IMU motion data. Users can inspect time-aligned signals and analyze movement characteristics within a single session.

- **Compare Sessions**

  Facilitates side-by-side comparison of multiple sessions, enabling comparative motion analysis across different recordings or subjects.

Together, these modules establish a structured workflow from data acquisition to advanced analysis, ensuring both usability and analytical robustness.

3.3 Signal Processing

In order to remove high-frequency noise from the IMU acceleration data, a first-order digital low-pass filter with a cut-off frequency of 10 Hz was applied. To eliminate low-frequency biases, such as gravitational acceleration and sensor drift, a high-pass filter with a cut-off frequency of 0.5 Hz was also applied. Filtering was only applied to the acceleration data, while angular velocity and orientation measurements were used without additional processing. The cut-off frequencies were chosen arbitrarily through trial and error. Furthermore, the filtering was applied to each spatial direction of the acceleration, and then the magnitude of the total acceleration was gained using the filtered directional accelerations. The total acceleration was calculated using the L2 norm.

3.4 Sensor Fusion and IMU Angle Computation

IMU data is uploaded to the web application as CSV files containing timestamped accelerometer, gyroscope, and magnetometer readings across three axes. The application requires 9-DOF data, meaning all three sensor modalities must be present. Prior to fusion, the magnetometer data is calibrated to remove hard iron bias and the accelerometer data is normalised to 1g before the filtered acceleration data from the signal processing stage is passed into the fusion pipeline.

Sensor fusion is performed using the Madgwick algorithm, implemented using a local reference copy of the AHRS: Attitude and Heading Reference Systems JavaScript library (22). The algorithm operates on the full 9-DOF MARG sensor array and produces a quaternion orientation estimate at each sample point, computed iteratively at the measured sample rate with a fixed gain parameter of $\beta = 0.1$. Each quaternion is subsequently converted to Euler angles (roll, pitch, yaw) and a 3×3 rotation matrix for further analysis and visualisation.

Initial joint angle estimation is performed using three IMUs worn simultaneously and placed on the shoulder, elbow, and wrist respectively. The relative orientation between adjacent body segments is computed at each time step by taking the quaternion product of one segment's orientation with the conjugate of the adjacent segment's orientation. The angle of this relative rotation quaternion is then extracted as:

$$\theta = 2 \arccos(w_{rel}) \tag{1}$$

where w_{rel} is the scalar component of the relative quaternion. This yields continuous elbow and wrist flexion angle time series, from which summary statistics including mean angle, minimum, maximum, and range of motion are computed.

4 Results

4.1 Web Application Progress

The implemented MoveSync system successfully integrates video-based pose estimation and IMU sensor data within a unified web-based analysis environment. The modular architecture enabled reliable data import, structured session management, and synchronized multimodal visualization without requiring external software or server-side processing.

The session-based data model functioned as intended, allowing each recording (one video file with associated IMU CSV files) to be stored, retrieved, and analysed as a session (i.e., a single coherent unit). Uploaded datasets were consistently parsed and organized within the Library module, demonstrating stable handling of data inputs.

In the Session Viewer, temporal synchronization between video-derived motion data and IMU measurements was achieved through independent preprocessing followed by alignment within a shared time reference. To establish this alignment, the user manually marks a corresponding event in both the video and the IMU data, thereby defining a temporal offset between the two data streams. This offset serves as the basis for synchronization within the session. The resulting alignment enables coherent playback and joint inspection of motion signals, confirming that the separation of data streams does not compromise integrated analysis. Visual inspection of synchronized sessions demonstrated consistent temporal correspondence between kinematic events observed in the video and variations in the IMU signals.

All processing was executed client-side within a standard web browser environment. Performance testing under typical dataset sizes showed that video loading, CSV parsing, synchronization, and visualization were completed without noticeable delays on commonly available hardware. This confirms the feasibility of a fully browser-based implementation for multimodal motion analysis.

The structured interface, consisting of Dashboard, Upload, Library, Session Viewer, and Compare Sessions modules, provided a coherent workflow from data acquisition to comparative analysis. The Compare Sessions module is currently under development and not yet fully implemented. While the interface structure and session selection logic have been established, full comparative visualization and analytical functionality remain subject to future work.

Overall, the results demonstrate that MoveSync fulfills its intended objectives: modular architecture, reliable multimodal synchronization, session-based data organization, and accessible browser-based deployment. The system establishes a functional foundation for further development, including advanced signal processing, automated feature extraction, and

quantitative performance metrics.

4.2 IMU Filtering

Figure 1 shows the raw (unfiltered) acceleration data recorded by the IMU. A constant bias of approximately ± 9.82 m/s^2 is visible on the z-axis, depending on the orientation of the sensor. This offset corresponds to the gravitational acceleration acting on the IMU. Furthermore, an acceleration of around ± 2 m/s^2 is shown on the y-axis. In addition, several sharp spikes can be observed in the acceleration signals.

Figure 2 presents the acceleration data after applying the filtering process. The filtered data indicates no bias and a smoother signal compared to the unfiltered data.

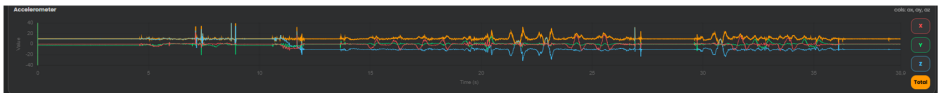


Figure 1: Unfiltered acceleration data from the IMU. The red line represents acceleration in the x-axis, the green line the y-axis, the blue line the z-axis and the yellow line the magnitude of the combined acceleration.

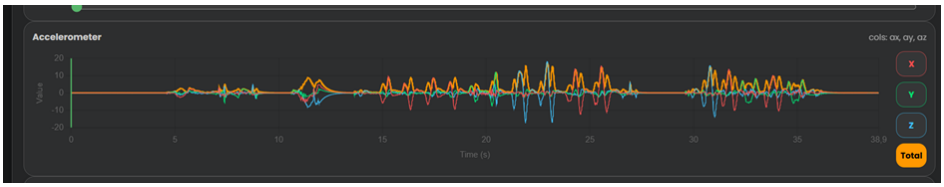


Figure 2: Filtered acceleration data from the IMU. The red line represents acceleration in the x-axis, the green line the y-axis, the blue line the z-axis and the yellow line the magnitude of the combined acceleration.

4.3 Sensor Fusion

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Figure 3 shows the sensor fusion output produced by the web application for a representative IMU session. The Madgwick algorithm successfully estimates the 9-DOF orientation of the sensor, displaying the result as a real-time 3D visualisation of the IMU body frame, alongside the corresponding quaternion, Euler angles, and rotation matrix at the selected time cursor position.

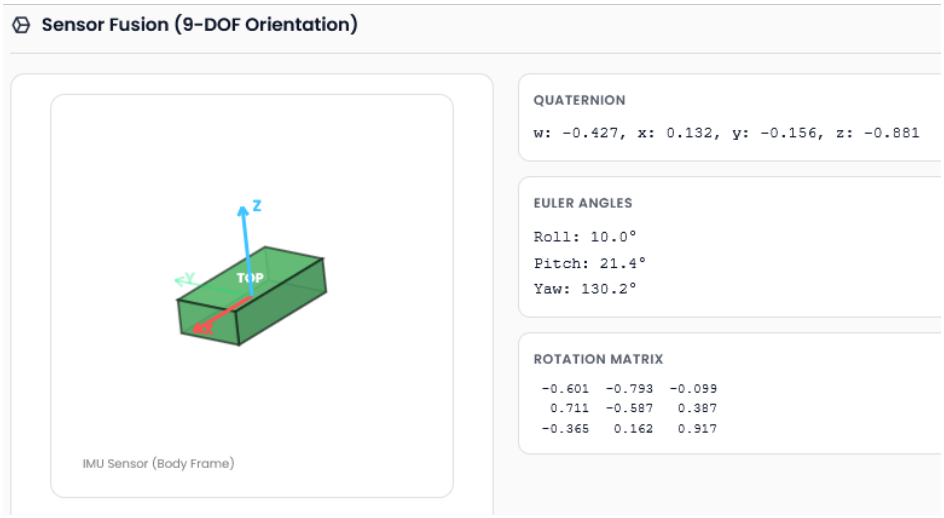


Figure 3: Web Application Sensor Fusion Output for a representative session.

4.4 IMU Angle Results

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Figure 4 shows the elbow and wrist angle time series computed from three simultaneously worn IMUs over a 60-second session. This currently has significant errors, with physically impossible angles and is something to be improved in future updates.

4.5 Future Experiments

One goal for the future work is to have working distance, velocity and acceleration calculations for the IMU data. To investigate the accuracy of these calculations we plan to setup a controlled space with set distances and compare manually calculated results with the output from our WebApp, with the analysis to go here.

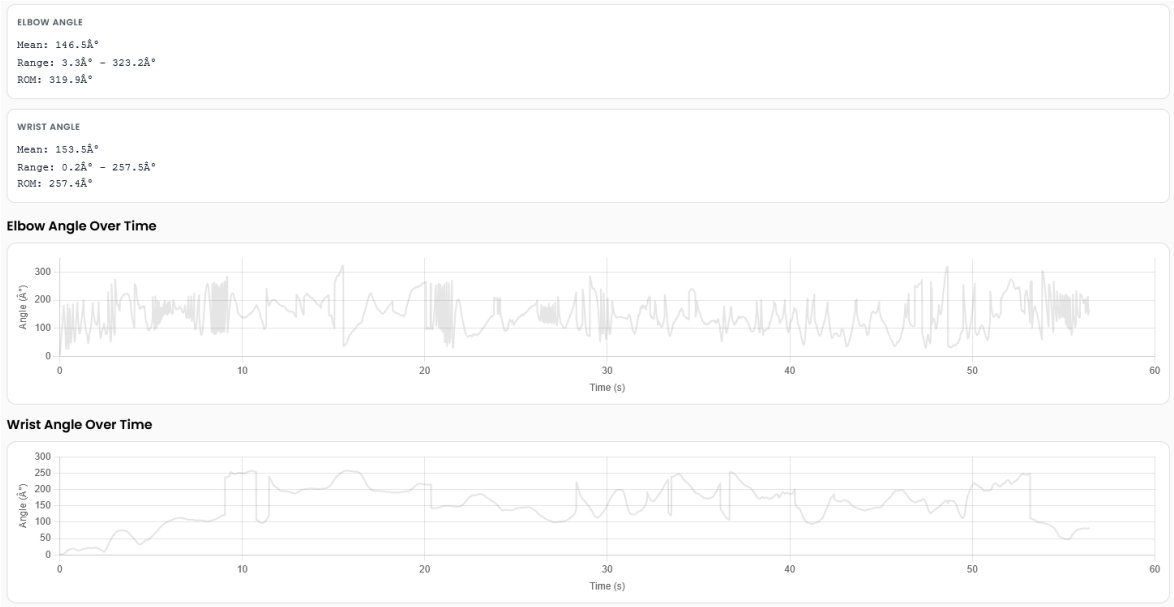


Figure 4: Elbow and wrist angle time series computed from three IMUs over a representative session.

5 Discussion

Write the discussion here.

6 Conclusion

Write the conclusion here.

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